Brittany Perera

brittanykawshalya@gmail.com

EXPERIENCE

Geospatial Engineer now

PangeAI

Working with satellite images and extracting analysis from them for hazards, risks and changes to the earth over time.

Powerplant Engineer & Software Developer 2024

Airline

- Built fleet-wide machine learning models that forecast engine deterioration and optimise maintenance across an airline fleet.
- Designed an interactive Powerplant Dashboard turning predictive models into real-time decision support for airline management.
- Modelled engine wash effectiveness and built wash scheduling strategies with measurable fuel and cost savings.

Undergraduate Researcher 2023 to 2024

Mercedes-AMG Petronas F1

- Optimised a radiator design for an F1 car using simulation and optimisation algorithms.
- Combined computational fluid dynamics with algorithmic design to improve cooling while cutting drag and mass.
- Built parametric simulation frameworks in MATLAB and Python to test designs across operating conditions.

EDUCATION

MEng Engineering Science 2020 to 2024

University of Oxford, St Edmund Hall

A-levels in Maths, Physics and Chemistry (3 As) 2013 to 2020

Torquay Girls' Grammar School

LEADERSHIP

College President 2021 to 2022

St Edmund Hall, Oxford

Represented 700+ students and led a 16-person committee.

May Ball President 2022

St Edmund Hall, Oxford

Organised a 950-person event with a 6-figure budget.

SKILLS

Python, SQL, MATLAB, C++, TypeScript, JavaScript, SolidWorks, CAD, scikit-learn, GitHub Actions

LANGUAGES

English (native), Sinhala (fluent), French (intermediate), German (intermediate)